Is Spurious Regression a Problem in Stationary Time Series?

Asymptotic Theory and Finite-Sample Monte Carlo Evidence

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Abstract

Regression between independent random walks is spurious: the OLS slope is inconsistent, and has a nondegenerate limit and the conventional t-ratio diverges. Stationarity removes that inconsistency, and it is tempting to conclude that conventional inference is then safe. This handout shows why that conclusion is wrong, and exactly how wrong. Under stationarity, ergodicity and weak dependence, OLS is consistent for the population projection slope, but the conventional t-ratio converges to $N(0, \lambda)$, where λ is the ratio of the long-run variance of the regression score to the product of the two variances the conventional formula uses. The test over-rejects, under-rejects, or is correctly sized according to whether λ exceeds, falls below, or equals one. We decompose λ into a contemporaneous and a serial component, which unifies three otherwise separate cases: independent AR(1) pairs, general independent linear processes, and dependent series with common volatility and a zero projection slope. Monte Carlo evidence covers nine data-generating processes, including a design in which both series are visibly serially correlated yet the conventional test is exactly correctly sized. Further experiments examine HAC inference, size-adjusted local power, local-to-unity sequences, and distributed-lag tests under omitted dynamics.

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1 Objectives and Intended Audience

1.1 Purpose

The question in the title is one that almost every empirical economist has met in an informal form: *my series are stationary, so am I safe?* The usual answer given in coursework is that spurious regression is a unit-root problem, that differencing or cointegration analysis disposes of it, and that stationary data may therefore be regressed in the ordinary way. That answer is incomplete in a way that matters for published work.

The objective of this handout is to replace it with a precise one. Specifically, the handout aims to establish three things: that stationarity secures the *estimate* but says nothing about the *standard error*; that whether conventional inference is trustworthy is settled by a single quantity, λ, which can be computed for any given data-generating process; and that λ can go wrong in two independent ways, only one of which is detected by the diagnostics most researchers actually run. A test for serial correlation is not sufficient, and the handout constructs explicit counterexamples in both directions — data that are visibly serially correlated yet require no correction at all, and data with no serial correlation whatsoever that reject a true null more than twice as often as they should.

The treatment is deliberately self-contained. Every claim is stated as a formal result with its assumptions written out, explained again in words, and then verified by simulation whose code is supplied and modifiable. Nothing is asserted that is not either proved or demonstrated numerically.

1.2 Two audiences

This handout is written for two overlapping groups, who will use it differently.

Readers building towards econometric theory and teaching. For staff members undertaking doctoral work with a view to specialising in econometric theory, and to teaching the subject at MA and PhD level, the material is intended as a compact case study in how asymptotic arguments are actually constructed and where they fail. The topic is well chosen for that purpose: it is elementary enough to be worked through completely, yet it involves a genuine limit theorem, a non-trivial variance calculation, a distinction between pointwise and uniform approximation, and a place where an intuitively appealing claim turns out to be false. Readers in this group should work through the proofs, attempt the exercises, and modify the simulation code — deriving λ for a process of their own choosing and confirming it numerically is the single most useful thing they can do with this handout.

Those preparing to teach may also find the *structure* of the handout worth borrowing independently of its content. Each formal result is followed by a **Logic of the argument** block giving the skeleton of the proof, and an **Intuition** block giving its meaning in ordinary language. Separating the three registers — what is claimed, why it is true, what it means — lets students who cannot yet follow the algebra still follow the economics, and lets those who can follow the algebra see what it is for. The device is reusable in any technical course.

Readers applying econometrics in their own empirical research. For academic staff whose concern is the credibility of applied work, the objective is narrower and more immediate: to be able to look at a regression on time-series data and judge whether the reported standard errors mean what they appear to mean. The practical content is concentrated in the Intuition blocks, in the results of Section 6.1 onwards, and in the final discussion. Readers in this group may skip every proof without loss. What they should not skip is the decomposition in Section 4.3, because it supplies the two questions that ought to be asked of any time-series regression, and Section 4.8, because it explains why passing a unit-root test is weaker reassurance than it appears.

1.3 What a reader should be able to do afterwards

On completing the handout, a reader should be able to:

1. state precisely what stationarity does and does not guarantee about OLS inference, and explain why the random-walk case is a different problem in kind rather than a more severe version of the same one;
2. compute λ for a given pair of processes, and recognise from the autocorrelation structure alone whether conventional inference will over-reject, under-reject, or be correct;
3. distinguish the two independent sources of failure — contemporaneous and serial — and identify which diagnostics detect which;
4. choose sensibly among conventional, HC and HAC standard errors, and say what each does not fix;
5. explain why a process that is technically stationary may still be badly approximated by stationary asymptotics, and why this makes near-unit-root diagnostics complementary to robust inference rather than redundant;
6. separate a specification failure from a covariance-estimation failure in a distributed-lag or Granger-causality setting, and know that only the second can be repaired by a robust covariance matrix; and
7. reproduce and extend the accompanying Monte Carlo evidence.

1.4 Prerequisites and technical level

The handout assumes familiarity with ordinary least squares, the notions of consistency and asymptotic normality, and the idea of an autocorrelation function. It does not assume measure-theoretic probability, and no result below requires it. Where a limit theorem is needed it is stated as an assumption in verifiable form and attributed, rather than proved from primitives; the mathematics that is carried out in full is confined to variance calculations that require no more than algebra and the summation of a geometric series.

Readers who prefer to see the argument before the formalism may read the Intuition blocks in Section 4 in sequence, ignoring everything between them, and will obtain a complete if informal account of the entire result.

2 Introduction

The problem is older than it is usually made to appear. Yule (1926) coined the term *nonsense-correlation* and, in what is essentially the first Monte Carlo study in econometrics, drew repeated samples to trace the distribution of the correlation coefficient across three classes of series. For independent random series it concentrated near zero, as expected. For *conjunct* series — serially correlated, but with differences that were not — the distribution was noticeably more dispersed. For conjunct series whose differences were also conjunct, which is essentially the integrated case, it became U-shaped, with most of its mass near ± 1: his motivating example, a correlation of 0.95 between the Church of England marriage rate and the standardised mortality rate in England and Wales over 1866–1911, was no accident of that particular pair of series but the typical outcome for such data.

Both of Yule's pathological classes reappear below. The U-shaped case anticipates, in simulation, the nondegenerate limit distribution later derived analytically; the intermediate *conjunct* case is the subject of this handout, and Yule's observation that mere serial correlation widens the sampling distribution is precisely the phenomenon quantified in Section 4.3 by the ratio λ.

Granger and Newbold (1974) returned to the question and showed that a regression between two independent random walks can produce a high R^2, a low Durbin–Watson statistic, and apparently significant coefficients even though the series are unrelated. Phillips (1986) then established the nonstandard asymptotic theory: the OLS slope has a nondegenerate Brownian-functional limit rather than converging to zero, while its conventional t-ratio diverges and the probability of a false rejection approaches one. Yule's U-shaped histogram and Phillips's limiting functional are, sixty years apart, descriptions of the same object.

It is tempting to infer that stationarity makes conventional regression inference harmless. That inference is too broad. For jointly stationary series with a zero population projection slope, OLS is consistent for zero under standard regularity conditions, but the conventional OLS standard error need not estimate the slope's long-run sampling variance. Full independence is sufficient but unnecessary: cross-lag dependence, common volatility, and other nonlinear dependence can coexist with a zero contemporaneous projection. The resulting test may over-reject, under-reject, or have correct asymptotic size. Which case occurs is determined by the long-run variance of the regression score, not by serial correlation alone.

This handout develops that distinction and evaluates it by Monte Carlo simulation. The experiment includes equal positive AR coefficients, unequal positive coefficients, an opposite-sign pair, an i.i.d. benchmark, an AR/MA design in which both series are serially correlated yet the conventional test is exactly correctly sized, dependent series with common stochastic volatility, local-to-unity sequences, and independent random walks. It compares conventional OLS inference with heteroskedasticity-consistent (HC) and heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors (White 1980; Newey and West 1987), measures what HAC costs in size-adjusted local power, and uses distributed-lag tests to examine Granger noncausality and the consequences of omitting the dependent variable's dynamics.

The results here are not new. The stationary over-rejection phenomenon was documented by Granger, Hyung and Jeon (2001), and Ventosa-Santaulària (2009) surveys the wider spurious-regression literature. The purpose of this handout is to assemble the argument in one place, with the assumptions stated explicitly and the simulation code available for inspection and modification.

The principal conclusions are:

1. Under suitable stationarity, ergodicity, and weak-dependence conditions, OLS consistently estimates the population projection slope; under a zero-projection null it converges to zero.

2. The conventional t-ratio converges to $N(0, \lambda)$, where λ is a long-run variance ratio. Values above, below, and equal to one imply over-rejection, under-rejection, and correct asymptotic size, respectively.
3. λ decomposes into a contemporaneous component κ and a serial component. Both must be accounted for: a serially uncorrelated score does *not* by itself deliver correct size, and a serially correlated score does not by itself destroy it.
4. For independent AR(1) series, λ depends on the product $\phi_x \phi_y$. Equal positive persistence produces over-rejection, whereas opposite-sign persistence can produce severe under-rejection.
5. HAC inference is asymptotically valid under additional conditions ensuring consistent long-run variance estimation, although highly persistent processes can exhibit substantial finite-sample distortion. Once critical values are size-adjusted, HAC costs remarkably little power.
6. Stationary asymptotics are not uniform near the unit-root boundary, and dynamic misspecification can generate false distributed-lag and Granger-causality findings even when the candidate predictor is generated independently.

3 The Classical Random-Walk Benchmark

Consider two independent random walks,

$$x_t = x_{t-1} + \varepsilon_t, \qquad y_t = y_{t-1} + \eta_t,$$

with independent zero-mean i.i.d. innovations of finite variance, and regress y_t on an intercept and x_t. Let $\bar{W}_j = \int_0^1 W_j(r)\,dr$. Functional central limit theory gives

$$\hat{\beta} \Rightarrow \frac{\int_0^1 \{W_x(r) - \bar{W}_x\}\{W_y(r) - \bar{W}_y\}\,dr}{\int_0^1 \{W_x(r) - \bar{W}_x\}^2\,dr}.$$

Thus $\hat{\beta} = O_p(1)$ and does not converge to the true value zero. The conventional standard error is $O_p(T^{-1/2})$, so the absolute t-ratio diverges at rate $T^{1/2}$ and

$$\Pr(|t| > 1.96) \longrightarrow 1.$$

This random-walk benchmark is the classical Granger–Newbold form of spurious regression. The analysis below contrasts it with fixed-parameter, short-memory stationary processes; it does not claim that unit roots are the only possible source of nonstandard regression asymptotics.

Intuition. A random walk wanders. Over any finite stretch it drifts up or down and traces out something that looks like a trend, even though nothing is pushing it in either direction. Regressing one random walk on another is therefore not really estimating a relationship between two variables; it is comparing the shape of one wandering path with the shape of another. Whether the two happen to drift the same way is a matter of luck in the particular pair of paths you drew.

Collecting more data does not settle the matter, because each new observation adds another step of wandering rather than another independent piece of evidence about a fixed relationship. That is why $\hat{\beta}$ never converges: there is nothing for it to converge to. The t-statistic grows without bound because the formula behind it counts every observation as fresh evidence, so with $T = 1000$ it believes it has a thousand times more information than it really does.

4 Stationary Regression Theory

4.1 Setup and regularity conditions

Consider the population linear projection

$$y_t = \alpha_0 + \beta_0 x_t + u_t, \qquad t = 1, \dots, T,$$

where

$$\beta_0 = \frac{\operatorname{Cov}(x_t, y_t)}{\operatorname{Var}(x_t)},$$

and test $H_0 : \beta_0 = 0$. Write $\mu_x = \mathbb{E}x_t$, $\mu_y = \mathbb{E}y_t$, $\sigma_x^2 = \operatorname{Var}(x_t) > 0$, $\sigma_u^2 = \operatorname{Var}(u_t)$, and define the **regression score**

$$w_t = (x_t - \mu_x)u_t.$$

Assumption 1 (Stationarity, moments, and projection orthogonality)**.** The joint process $\{(x_t, y_t)\}$ is strictly stationary and ergodic, has finite moments of order $4 + \delta$ for some $\delta > 0$, and satisfies the population projection condition $\mathbb{E}[(x_t - \mu_x)u_t] = 0$. Under H_0, $u_t = y_t - \mu_y$.

Assumption 2 (Weak dependence and central limit theorem)**.** The joint process is sufficiently weakly dependent that

$$T^{-1/2}\sum_{t=1}^{T} w_t \Rightarrow N(0, \Omega), \qquad \Omega = \sum_{k=-\infty}^{\infty} \operatorname{Cov}(w_t, w_{t+k}) \in (0, \infty),$$

and the autocovariances required below are absolutely summable.

Standard strong-mixing conditions with suitable moment and mixing-rate restrictions are sufficient for Assumption 2 (Davidson 1994, 2000).

Assumption 3 (HAC consistency)**.** For HAC inference, a positive-semidefinite kernel and a bandwidth L_T are used under conditions ensuring uniform consistency of the sample autocovariances, with $L_T \to \infty$ and $L_T/T \to 0$. These conditions imply $\widehat{\operatorname{LRV}}(w) \xrightarrow{p} \Omega$.

Primitive conditions for Assumption 3 are given by Andrews (1991). Note that it is additional to the score central limit theorem: a CLT alone does not guarantee that a particular HAC estimator consistently estimates Ω.

Intuition (what the three assumptions are each for)**.** Read the three assumptions as a division of labour, each buying one thing.

Assumption 1 says the process does not change its character over time and that averages taken from it settle down as the sample grows. This is what makes the *estimate* reliable. The moment condition is bookkeeping: we will be averaging products of four random quantities, so those products need to have a finite average in the first place.

Assumption 2 says that a particular average — the average of the score w_t — is not just settling down but is approximately normally distributed, with a spread measured by Ω. This is what makes *testing* possible at all, since every critical value we use comes from a normal distribution. The quantity Ω is the true spread of that average.

Assumption 3 says we can *estimate* Ω from the data. This is a genuinely separate requirement and it is the one most often assumed away. Knowing that a quantity has a normal distribution with some spread is not the same as being able to measure that spread from a finite sample. Assumption 2 gives us the first; only Assumption 3 gives us the second.

4.2 Consistency and the conventional test

Proposition 4 (Consistency)**.** *Under Assumption 1,*

$$\hat{\beta} \xrightarrow{p} \beta_0,$$

and hence $\hat{\beta} \xrightarrow{p} 0$ *under* H_0.

Proof. The OLS slope is the ratio of the sample covariance of x_t and y_t to the sample variance of x_t. By ergodicity, these quantities converge to $\mathrm{Cov}(x_t, y_t)$ and $\sigma_x^2 > 0$, respectively. Their ratio is β_0; under the null it is zero. ■

Intuition. The estimated slope is built from two averages: an average measuring how x and y move together, and an average measuring how much x moves on its own. If both averages settle down to the right numbers, then so does their ratio, and that is the whole proof.

Notice how little is being asked. Nothing here rules out serial correlation; the data may be strongly dependent from one period to the next. Dependence makes the averages settle down more slowly, so you need a longer sample to get close, but it does not stop them settling. This is exactly what fails for random walks: their averages never settle at all, which is why the random-walk case is a different problem in kind rather than a more severe version of the same one.

The practical message is a narrow one, and it is easy to over-read. Stationarity protects the *number* you report as your estimate. It says nothing whatsoever about the standard error printed next to it, which is the subject of the rest of this section.

Proposition 5 (Slope and conventional t-ratio)**.** *Under Assumptions 1–2 and* H_0*,*

$$\sqrt{T}\hat{\beta} \Rightarrow N\left(0, \frac{\Omega}{\sigma_x^4}\right)$$

and the conventional OLS t-ratio satisfies

$$t_{OLS} \Rightarrow N(0, \lambda), \qquad \lambda = \frac{\Omega}{\sigma_x^2 \sigma_u^2} = \frac{\mathrm{LRV}\{(x_t - \mu_x)u_t\}}{\mathrm{Var}(x_t)\mathrm{Var}(u_t)}.$$

Proof. Demeaning changes the scaled score numerator by $o_p(1)$, while $T^{-1}\sum_t (x_t - \bar{x})^2 \xrightarrow{p} \sigma_x^2$. Assumption 2 and Slutsky's theorem give the slope limit. The conventional standard error obeys

$$T\, \mathrm{se}_{OLS}(\hat{\beta})^2 \xrightarrow{p} \frac{\sigma_u^2}{\sigma_x^2},$$

which yields the stated t-ratio limit. ■

For a two-sided nominal 5% test,

$$\Pr(|t_{OLS}| > 1.96) \longrightarrow 2\Phi\left(-\frac{1.96}{\sqrt{\lambda}}\right).$$

Therefore $\lambda > 1$ implies asymptotic over-rejection, $\lambda < 1$ implies asymptotic under-rejection, and $\lambda = 1$ implies correct asymptotic size.

An equivalent and often more useful statement: the *size-adjusted* two-sided critical value for the conventional statistic converges to $1.96\sqrt{\lambda}$. Section 6.3 verifies this directly in simulation, which is a sharper test of Proposition 5 than any tail rejection probability.

Logic of the argument. A t-statistic is an estimate divided by an estimate of its own spread. Proposition 5 works out each piece separately and then compares them.

The *true* spread of the slope estimate is governed by Ω, the spread of the summed score. The *assumed* spread, the one the conventional formula computes, is $\sigma_x^2\sigma_u^2$ — the variance of the regressor multiplied by the variance of the error, as if the two could be handled separately and as if time-ordering were irrelevant.

If those two quantities agree, the t-statistic has the standard normal distribution we all assume it has. If they disagree, the statistic is still normal and still centred at zero, but its spread is wrong by the factor λ, the ratio of truth to assumption. Everything else in this handout is an investigation of when that ratio is not one.

Intuition. The conventional standard error assumes that every observation is a fresh piece of information. When the data are dependent, that is false, and the formula misprices the evidence.

Suppose consecutive observations tend to repeat each other. Then a sample of 200 does not carry 200 independent pieces of information — perhaps it carries the equivalent of 50. The conventional formula still divides as though there were 200, producing a standard error that is too small, a t-statistic that is too large, and rejections far more often than 5% of the time. In this reading λ is a penalty for dependence, and T/λ is the *effective* sample size: with $\lambda = 4$, a sample of 200 is worth about 50.

The reverse is equally possible and much less well known. If consecutive observations tend to offset each other, the sample carries *more* information than its length suggests, the conventional standard error is too large, and the test rejects too rarely. Such a test is not "safe" — it will miss relationships that are really there. Section 6.1 reports a case that rejects essentially never.

So $\lambda > 1$ means "you have less evidence than you think", $\lambda < 1$ means "you have more", and $\lambda = 1$ means the conventional formula happens to be right. Note that a t-statistic of 2.5 is not evidence of anything by itself: with $\lambda = 4$ the honest cutoff is $1.96 \times 2 = 3.92$, so the same number that looks convincing under one λ is unremarkable under another.

4.3 What makes λ differ from one

The following decomposition is the organising device for everything that follows. It is elementary, but it prevents a mistake that is easy to make.

Lemma 6 (Contemporaneous and serial components)**.** *Under Assumptions 1–2,*

$$\lambda = \underbrace{\frac{\mathbb{E}[(x_t-\mu_x)^2u_t^2]}{\sigma_x^2\sigma_u^2}}_{\kappa} + \underbrace{\frac{2}{\sigma_x^2\sigma_u^2}\sum_{k=1}^{\infty}\operatorname{Cov}(w_t,w_{t+k})}_{\textit{serial component}} .$$

Proof. $\mathbb{E}w_t = 0$ by Assumption 1, so $\operatorname{Cov}(w_t,w_t) = \mathbb{E}[(x_t-\mu_x)^2u_t^2]$. Split $\Omega = \sum_k \operatorname{Cov}(w_t,w_{t+k})$ into its lag-zero term and the symmetric sum over $k \neq 0$, then divide by $\sigma_x^2\sigma_u^2$. ■

The quantity κ measures contemporaneous dependence between the squared regressor deviation and the squared error: $\kappa = 1$ when $\mathbb{E}[(x_t-\mu_x)^2u_t^2]$ factorises, and $\kappa \neq 1$ under conditional heteroskedasticity. The serial component measures the autocovariance of the score.

Logic of the argument. Ω is a sum of terms, one for each possible gap between two time periods. The lemma does nothing more than split that sum into the single term with no gap and everything else, then divide through so that each part is expressed relative to what the conventional formula assumes.

Intuition. This is the most useful thing in the handout, so it is worth stating in words.

There are exactly two ways the conventional standard error can be wrong, and they have nothing to do with each other.

The first is a failure *at a single point in time*, measured by κ. The conventional formula multiplies the typical size of the regressor by the typical size of the error, treating them as unrelated. Suppose instead that periods with unusually large x tend to be the same periods with unusually large errors. Then the products $x_t u_t$ have occasional very large values, the sum of them is far more volatile than the formula expects, and κ exceeds one. Nothing about the ordering of the data is involved — shuffle the observations into random order and this problem survives untouched.

The second is a failure *across time*, measured by the serial component. Here each product $x_t u_t$ may be perfectly well behaved on its own, but consecutive products tend to have the same sign, so they accumulate instead of cancelling. Shuffle the data and this problem disappears entirely.

The mistake this handout is written to prevent is checking only the second. A researcher who tests for serial correlation, finds none, and concludes the conventional standard error is safe has checked one of the two terms and ignored the other. Section 4.7 gives a process where the serial term is exactly zero and the test still rejects two and a half times too often.

A useful mental image: κ asks whether the trouble is visible in a scatter plot, and the serial term asks whether it is visible in a time plot. You have to look at both.

Remark 7 (The condition for correct size). Correct asymptotic size requires $\lambda = 1$, that is, $\Omega = \sigma_x^2 \sigma_u^2$. This needs *both* components to be accounted for. In particular:

- A serially uncorrelated score is **not** sufficient. If the serial component vanishes then $\lambda = \kappa$, and $\kappa = 1$ only under score conditional homoskedasticity. Section 4.7 constructs a process whose score is exactly serially uncorrelated with $\kappa = e^{v_h} > 1$.
- A serially correlated score is **not** fatal. Positive and negative autocovariances can cancel, leaving $\lambda = 1$ with both series visibly autocorrelated. Section 4.6 constructs such a case and Section 6.1 confirms by simulation that it is correctly sized.

A sufficient condition for $\lambda = 1$ is that w_t be serially uncorrelated *and* u_t be independent of x_t — for example, u_t i.i.d. and independent of the regressor.

Remark 8 (Exact Gaussian benchmark). If the regression errors are i.i.d. Gaussian and independent of the complete regressor matrix, the usual t-statistic has an exact Student-t distribution conditional on the regressors. The i.i.d. Gaussian Monte Carlo DGP below satisfies these stronger conditions. Without them, $\lambda = 1$ is an asymptotic statement and does not imply exact finite-sample size.

4.4 Independent AR(1) processes with unequal coefficients

Proposition 9 (General AR(1) example)**.** *Let*

$$x_t = \phi_x x_{t-1} + \varepsilon_t, \qquad y_t = \phi_y y_{t-1} + \eta_t,$$

where $|\phi_x| < 1$, $|\phi_y| < 1$, and $\{\varepsilon_t\}$ and $\{\eta_t\}$ are mutually independent i.i.d. sequences with zero mean and finite fourth moments. Then $\kappa = 1$ and

$$\lambda = \frac{1 + \phi_x \phi_y}{1 - \phi_x \phi_y}.$$

Proof. Independence of the two processes gives

$$\mathrm{Cov}(w_t, w_{t+k}) = \gamma_x(k)\gamma_y(k) = (\phi_x \phi_y)^{|k|} \sigma_x^2 \sigma_y^2.$$

At $k = 0$ this is $\sigma_x^2\sigma_y^2$, so $\kappa = 1$ by Lemma 6. Summing over all lags,

$$\Omega = \sigma_x^2\sigma_y^2 \left\{ 1 + 2\sum_{k=1}^{\infty} (\phi_x\phi_y)^k \right\} = \sigma_x^2\sigma_y^2 \frac{1 + \phi_x\phi_y}{1 - \phi_x\phi_y}. \quad \blacksquare$$

Note that Gaussianity is not required: mutual independence of the two innovation *processes* plus finite fourth moments is enough. Equal positive coefficients reproduce the familiar over-rejection result. Unequal positive coefficients show that the relevant quantity is their product. Opposite signs make that product negative and can cause substantial under-rejection, despite serial correlation in both series.

Intuition (why the *product* of the two coefficients is what matters)**.** The score is the product $x_t y_t$. For that product to carry information from one period to the next, *both* series must carry information from one period to the next. If either series forgets its past immediately, the product forgets too.

This has a consequence worth memorising: a regression whose regressor is serially uncorrelated is safe no matter how badly autocorrelated the error is. Set $\phi_x = 0$ and $\lambda = 1$ for every value of ϕ_y. Textbook warnings about autocorrelated errors are really warnings about autocorrelated errors *together with* a persistent regressor.

The sign works the same way. If both series are positively autocorrelated, then a large x this period tends to be followed by a large x next period, and likewise for y, so consecutive products line up and reinforce one another: λ rises and the test over-rejects. If one series is positively and the other negatively autocorrelated, consecutive products tend to *flip* sign, so they cancel; with $(0.8, -0.8)$ the cancellation overshoots so far that λ falls to about 0.22 and the test almost never rejects.

The numbers escalate sharply. Moving from $\phi = 0.5$ to $\phi = 0.95$ in both series raises λ from 1.7 to 19.5, which means an effective sample size roughly twenty times smaller than the nominal one. Persistence of 0.95 is entirely ordinary in macroeconomic and financial data.

4.5 Independent linear processes

The AR(1) calculation extends naturally. Let

$$x_t = \sum_{j=0}^{\infty} a_j \varepsilon_{t-j}, \qquad y_t = \sum_{j=0}^{\infty} b_j \eta_{t-j},$$

where the coefficient sequences are absolutely summable and the two innovation sequences are mutually independent. If $\gamma_x(k)$ and $\gamma_y(k)$ denote their autocovariances and $\rho_x(k)$, $\rho_y(k)$ the corresponding autocorrelations, then independence implies $\kappa = 1$ and

$$\Omega = \sum_{k=-\infty}^{\infty} \gamma_x(k)\gamma_y(k), \qquad \lambda = \sum_{k=-\infty}^{\infty} \rho_x(k)\rho_y(k).$$

This formula covers short-memory ARMA processes, unequal lag structures, seasonal dependence, and negative autocorrelation. It also shows why the *sign pattern* of the two autocorrelation functions, rather than persistence in either series separately, governs conventional size.

Intuition. Each series has an autocorrelation function: a list of numbers recording how strongly it resembles itself one period back, two periods back, and so on. The formula says: line the two lists up side by side, multiply them lag by lag, and add everything up. That total is λ.

Read this way, λ measures the *alignment* of the two autocorrelation functions, not the size of either. Two series can each be strongly autocorrelated and still produce $\lambda = 1$, provided their patterns fail to line up — where one has a positive correlation the other has a negative one, and the contributions cancel. Equally, two mildly autocorrelated series that happen to share the same pattern will produce $\lambda > 1$.

So the question to ask about a regression is not "are my series persistent?" but "do my series resemble themselves over time *in the same way*?". Section 4.6 builds a pair that deliberately fails to line up, and the conventional test on that pair is exactly correct.

4.6 A correctly sized test with serially correlated data

The formula above makes the "not necessary" half of Remark 7 constructive. Take x_t to be AR(1) with coefficient ϕ, so $\rho_x(k) = \phi^{|k|}$, and y_t to be an invertible MA(2),

$$y_t = \eta_t + \theta_1\eta_{t-1} + \theta_2\eta_{t-2},$$

whose autocorrelations vanish beyond lag two. Then

$$\lambda = 1 + 2\left\{\phi\rho_y(1) + \phi^2\rho_y(2)\right\},$$

so $\lambda = 1$ exactly whenever $\rho_y(1) + \phi\rho_y(2) = 0$, which reduces to

$$\theta_1(1+\theta_2) + \phi\theta_2 = 0 \qquad\Longleftrightarrow\qquad \theta_1 = -\frac{\phi\theta_2}{1+\theta_2}.$$

The simulation uses $\phi = 0.5$ and $\theta_2 = 0.5$, hence $\theta_1 = -0.1667$. The implied autocorrelations are $\rho_x(1) = 0.500$, $\rho_y(1) = -0.196$ and $\rho_y(2) = 0.391$: both series are plainly serially correlated, and the MA roots have modulus $\sqrt{2} > 1$ so the process is invertible. Yet the two contributions cancel exactly and the conventional test is correctly sized. This is the cleanest available demonstration that serial correlation is not the diagnostic — the sign pattern is.

Intuition. We are building y so that its autocorrelation pattern deliberately fails to line up with x's.

The regressor x is a simple persistent series: it resembles itself with correlation 0.5 one period back and 0.25 two periods back, both positive. We then choose y so that it resembles itself *negatively* one period back and *positively* two periods back. When the two lists are multiplied lag by lag, the negative contribution at lag one and the positive contribution at lag two are equal and opposite, and they cancel to nothing.

The result is a pair of series that would fail any test for serial correlation you ran on them individually — a Durbin–Watson statistic on either would look alarming — yet the ordinary t-test on the regression between them is exactly right, with no correction of any kind needed. A student who had learned "autocorrelation means your standard errors are wrong" would apply a correction here and make the answer worse rather than better.

The cancellation is of course exact only because we engineered it. Real data will not oblige. The point is not that this configuration is common but that it is possible, which is enough to show that the presence of serial correlation cannot by itself tell you what to do.

4.7 Dependence without a contemporaneous projection

Full independence is stronger than the null $\beta_0 = 0$. For example, let

$$x_t = s_t\varepsilon_t, \qquad y_t = s_t\eta_t,$$

where $s_t > 0$ is a common persistent stochastic scale and the two innovations are mutually independent, serially independent, and independent of s_t. Then $\mathrm{Cov}(x_t, y_t) = 0$, but the series are dependent through their common volatility. The score $w_t = s_t^2\varepsilon_t\eta_t$ has $\mathrm{Cov}(w_t, w_{t+k}) = 0$ for every $k \neq 0$, so the serial component of Lemma 6 vanishes identically. If $h_t = \log s_t^2$ is Gaussian with variance v_h, then

$$\lambda = \kappa = \frac{\mathbb{E}(s_t^4)}{\mathbb{E}(s_t^2)^2} = e^{v_h} > 1.$$

This is the promised counterexample to “serially uncorrelated score implies correct size”. The distortion is entirely contemporaneous heteroskedasticity. An HC standard error is sufficient asymptotically; HAC is also valid but can be less precise in small samples.

Intuition. Think of s_t as the general level of turbulence in the market at time t. Both series are calm in calm periods and wild in wild periods, but which *direction* each moves is decided independently by its own coin flip. So the two variables are genuinely unrelated in the sense the null cares about — knowing x tells you nothing about the direction of y — while being obviously related in another sense, since knowing that x was wild tells you y was wild too.

Now consider the products $x_t y_t$ that the standard error is built from. In calm periods they are all tiny. In wild periods both factors are large at once, so the product is enormous. The sum is therefore dominated by a handful of turbulent episodes, which makes it far more volatile than a formula assuming steady conditions would predict. That formula multiplies the average size of x by the average size of y and never notices that the large values arrive *together*. Here the mismatch is a factor of about 1.65, and the test rejects roughly 12.7% of the time instead of 5%.

The decisive feature is what happens if you shuffle the observations into random order. Serial correlation would be destroyed by shuffling; this problem would not, because it lives entirely in the pairing of x and y within each period. That is why the correct remedy is HC, which looks at each period’s product on its own, rather than HAC, which is designed to detect relationships across periods that here do not exist.

The pattern is not exotic. Volatility clustering of exactly this kind is among the most robust regularities in financial and macroeconomic data.

4.8 Near-unit roots and non-uniformity

Fixed-parameter stationary theory holds $|\phi| < 1$ constant as T grows. It need not approximate sequences such as

$$\phi_T = 1 - \frac{c}{T}, \qquad c > 0,$$

which approach the unit-root boundary with the sample size. Local-to-unity asymptotics (Phillips 1987; Chan and Wei 1987) are nonstandard: $\hat{\beta}$ again has a nondegenerate limit — now a functional of independent Ornstein–Uhlenbeck processes rather than Brownian motions — and the t-ratio again diverges at rate $T^{1/2}$, so the rejection probability tends to one exactly as in the unit-root case. The fixed-ϕ plateau formula is therefore not uniform over such sequences. Consequently, classifying a fitted process as technically stationary does not guarantee that ordinary stationary or HAC approximations are accurate in realistic samples. This non-uniformity is the source of well-known difficulties in predictive regression (Cavanagh et al. 1995; Elliott 1998; Stambaugh 1999).

Intuition (what ”non-uniform” means, and why it is not a technicality)**.** Asymptotic theory is a way of producing approximations, not a description of what actually happens as you collect data. When we say ”let $T \to \infty$ with ϕ fixed at 0.99”, we are choosing to approximate our situation by one in which the sample is enormous relative to how close ϕ sits to one. Whether that is a good approximation depends on our actual sample.

For $\phi = 0.99$ and $T = 200$, it is a poor one. Such a series has barely had time to revert towards its mean even once, so within the sample it behaves for all practical purposes like a random walk. The local-to-unity device is a different approximation, built to describe exactly this situation: it lets ϕ approach one as T grows so that the two stay in a realistic relationship to each other.

”Non-uniform” is the precise statement of the difficulty. For *every* fixed ϕ below one, the ordinary stationary theory is eventually correct — that much is true and is not in dispute. But ”eventually” arrives later and later as ϕ approaches one, and there is no single sample size that is large enough for all $\phi < 1$ at once. So the reassurance ”my series is stationary, therefore stationary theory applies” is empty unless you also ask *how* stationary, relative to how much data you have.

The practical upshot is that a unit-root test which fails to reject is not a licence to proceed. An estimated $\hat{\phi}$ of 0.97 in 150 observations should be treated as a warning, not a clearance, whichever side of the critical value it falls.

4.9 HAC inference

Using the Bartlett kernel, define

$$\widehat{\mathrm{LRV}}(w) = \hat{\gamma}_0 + 2\sum_{k=1}^{L_T}\left(1 - \frac{k}{L_T+1}\right)\hat{\gamma}_k, \qquad \hat{\gamma}_k = T^{-1}\sum_{t=k+1}^{T}\hat{w}_t\hat{w}_{t-k},$$

where $\hat{w}_t = (x_t - \bar{x})\hat{u}_t$.

Proposition 10 (HAC validity)**.** *Under Assumptions 1–3,*

$$t_{HAC} = \frac{\hat{\beta}}{\mathrm{se}_{HAC}(\hat{\beta})} \Rightarrow N(0,1).$$

The proof is standard and is omitted (Newey and West 1987; Andrews 1991).

This result is asymptotic, and three caveats matter in practice. First, when persistence is high a finite sample may contain too little information to estimate the long autocovariance tail accurately, so bandwidth choice is consequential even though any valid bandwidth sequence delivers consistency. Second, prewhitening (Andrews and Monahan 1992) often improves matters substantially in exactly the persistent cases where the plain estimator struggles. Third, the $N(0,1)$ limit ignores the sampling variability of $\widehat{\mathrm{LRV}}$; fixed-b asymptotics (Kiefer and Vogelsang 2005) retains it and delivers larger, better-calibrated critical values. The small-sample HAC over-rejection documented in Section 6.2 is the canonical motivation for that literature; Lazarus et al. (2018) give practical recommendations.

Logic of the argument. Only one piece of the conventional t-statistic is broken. The estimate in the numerator is fine, by Proposition 4. The denominator is wrong because it uses $\sigma_x^2\sigma_u^2$ where it should use Ω. So replace it: estimate Ω directly from the data and divide by that instead. If the estimate is any good, the resulting statistic has the standard normal distribution the conventional one was supposed to have.

Intuition (why the bandwidth is a genuine dilemma)**.** To estimate Ω we must estimate how strongly the score resembles itself one period back, two periods back, and so on — and then add all of those up. In principle the list is infinite; in practice we can only use a finite number of terms, and L_T is how many we use.

Both directions fail, which is what makes this awkward. Use too few lags and you miss the tail of the list, understate Ω, and remain oversized — you have corrected part of the problem and declared victory. Use too many and each individual term is estimated from very little data: the correlation at lag 40 in a sample of 200 rests on 160 overlapping pairs and is mostly noise. Adding up many noisy numbers gives a noisy total, and a standard error that is itself unreliable produces unreliable tests. There is no setting that avoids both errors; the bandwidth rule is a compromise.

This is why HAC is not a switch you flip. When $\phi = 0.95$ the score's memory is so long that no bandwidth a sample of even 5000 can support captures the whole tail, which is exactly what the HAC comparison in Section 6.2 shows: HAC still rejects about twice as often as it should. It is also why HAC can be *worse* than doing nothing at very small T — at $T = 50$ even the independent-data benchmark, where there is nothing whatever to correct, over-rejects under HAC simply because the correction itself is estimated noisily.

4.10 Distributed lags, Granger causality, and omitted dynamics

A distributed-lag regression asks whether past values of x_t contain predictive information for y_t. In the dynamic model

$$y_t = \alpha + \sum_{j=1}^{p} \rho_j y_{t-j} + \sum_{j=1}^{q} \beta_j x_{t-j} + e_t,$$

x does not Granger-cause y relative to the stated information set when $H_0 : \beta_1 = \cdots = \beta_q = 0$ (Granger 1969). This is a predictive definition, not a claim about structural causality.

Correct specification matters as much as the covariance estimator. If relevant lags of y_t are omitted, their persistence enters the regression error. Persistent x lags multiplied by that serially correlated error produce a dependent score vector, so the conventional joint Wald test can falsely detect predictive content.

It is worth being precise about *which* failure this is, because two distinct problems are easily conflated. Omitting a relevant regressor biases the remaining coefficients only if the omitted variable is correlated with the included ones. In the experiment of Section 6.6 the two series are generated independently, so the omitted y_{t-1} is orthogonal to x_{t-1} and x_{t-2}: the population projection coefficients on the x lags remain exactly zero, there is no bias, and the entire distortion is a failure of the conventional covariance estimator. That is why HAC repairs most of it. In applications where the omitted dynamics *are* correlated with the included regressors, the coefficients are biased as well, and no covariance estimator can help. Including an adequate dynamic specification remains the first line of defence; robust covariance estimation is not a substitute for specifying the conditional mean.

Intuition (two different failures that look alike in the output)**.** Leaving a variable out of a regression can go wrong in two quite separate ways, and students routinely merge them into one.

The first is *bias*: the estimate itself moves to the wrong place. This happens only when the omitted variable is correlated with the variables you kept, so that the regression credits the missing variable's influence to whichever included variable resembles it. No standard error can repair this, because the number being tested is already wrong.

The second is a *mismeasured standard error*: the estimate is fine, but the uncertainty attached to it is not. This is what happens when you omit something uncorrelated with your regressors. Its influence goes into the error term, which becomes persistent, and persistent errors combined with a persistent regressor inflate λ exactly as in Section 4.3.

The experiment in Section 6.6 is deliberately of the second kind: x and y are built independently, so dropping y_{t-1} cannot bias anything. Everything you see there — a test rejecting a true null about a third of the time — comes purely from a mismeasured standard error, which is why HAC repairs most of it. Do not read the experiment as showing that HAC fixes omitted-variable bias. It does not, and in applications where the omitted dynamics *are* correlated with your regressors you have both problems and HAC addresses only one.

The useful diagnostic question is therefore: is the thing I left out correlated with the things I kept? If no, you have a standard-error problem and a robust covariance estimator is the right tool. If yes, you have a specification problem and no covariance estimator will save you.

Theorem 11 (Summary of the stationary case)**.** *For fixed-parameter, short-memory stationary series satisfying Assumptions 1–2, OLS is consistent for the population projection slope. Under $H_0 : \beta_0 = 0$, conventional inference is governed by $\lambda = \kappa +$ serial component. The test over-rejects when $\lambda > 1$, under-rejects when $\lambda < 1$, and has correct asymptotic size when $\lambda = 1$. Under the additional HAC-consistency conditions in Assumption 3, HAC inference has correct asymptotic size. Thus stationarity prevents the random-walk inconsistency considered here, but does not by itself validate conventional OLS standard errors.*

What to take away. Stationarity buys you consistency of $\hat{\beta}$, and nothing else. Whether the conventional t-test is trustworthy is a separate question, answered by a single number λ, which has two independent sources of trouble: contemporaneous heteroskedasticity of the score ($\kappa \neq 1$) and serial correlation of the score. Neither serial correlation in the data nor its absence tells you what λ is. Diagnose the score, not the series.

5 Monte Carlo Design

The simulation uses independent Gaussian innovations and nine DGPs:

1. an i.i.d. Gaussian benchmark;
2. equal positive AR coefficients $(\phi_x, \phi_y) = (0.5, 0.5)$, $(0.8, 0.8)$, $(0.9, 0.9)$, and $(0.95, 0.95)$;
3. unequal positive coefficients $(0.9, 0.5)$;
4. opposite-sign coefficients $(0.8, -0.8)$;
5. the AR/MA pair of Section 4.6, for which $\lambda = 1$ exactly despite serial correlation in both series; and
6. independent random walks as a nonstationary benchmark.

For every stationary AR(1) process, the pre-sample state is drawn independently from its invariant distribution $N\{0, 1/(1-\phi^2)\}$; the MA(2) process is stationary from its first observation. Every retained observation is therefore drawn from the stationary process. One trajectory of length 5000 is generated in each replication, and nested prefixes are evaluated on the grid

$$T \in \{50, 100, 150, \dots, 5000\}.$$

Nesting correlates adjacent estimates and produces a smoother comparison across T; it does not eliminate Monte Carlo sampling variation.

The experiment uses $N = 2000$ replications per DGP. The maximum Monte Carlo standard error of a rejection proportion is $\sqrt{0.25/2000} \approx 1.12$ percentage points, and reported proportions should be read with that in mind. The conventional statistic is evaluated over the complete grid. HAC statistics are evaluated at $T \in \{50, 200, 500, 1000, 2000, 5000\}$ using a Bartlett kernel and $L_T = \lfloor 2T^{1/3} \rfloor$, which satisfies $L_T \to \infty$ and $L_T/T \to 0$. This rate is the MSE-optimal one for the Bartlett kernel and is deliberately more generous than the Newey–West rule of thumb $4(T/100)^{2/9}$, because the scores generated by persistent AR(1) pairs have a long autocovariance tail; Section 6.2 shows that even this bandwidth is not generous enough at $\phi = 0.95$.

The standalone script `scripts/spurious_stationary_sim.R` is the single computational source for the document. It generates all data, figures, and CSV tables, and every numerical value quoted in the text below is read from its output at render time rather than transcribed by hand. A single Bartlett estimator, implemented in that script, is used for every HAC number reported here, so the tables are mutually comparable.

Four supplementary experiments broaden the design:

1. **Common stochastic volatility.** A persistent Gaussian log-volatility with autoregressive coefficient 0.95 and stationary variance 0.5 multiplies independent Gaussian innovations in both series. The projection slope is zero, but the series are dependent and the theoretical variance ratio is $e^{0.5} \approx 1.65$. Conventional, HC0, and HAC inference are compared using 2000 replications.
2. **Local-to-unity sequences.** At each selected T, independent AR(1) series use $\phi_T = 1 - 5/T$. Because the DGP changes with T, this is a triangular-array experiment rather than a nested

fixed-parameter trajectory. The pre-sample value is drawn from the invariant distribution, so this is the *stationary-start* design; a zero-start design gives materially different finite-sample rejection frequencies, and the initialisation is a modelling choice rather than a technicality. Conventional and HAC tests use 1000 replications.

3. **Distributed-lag Granger tests.** Independent AR(1) series with $(\phi_x, \phi_y) = (0.8, 0.8)$ are used to test jointly whether two lags of x predict y. The correctly specified regression includes y_{t-1}; the misspecified regression omits it. Conventional and HAC Wald tests are compared over 1000 replications at $T \in \{100, 250, 500, 1000, 2000\}$.
4. **Size-adjusted local power.** Under $y_t = (b/\sqrt{T})x_t + v_t$ with x and v independent AR(1) processes with $\phi = 0.8$, conventional and HAC tests are compared at $T \in \{250, 1000\}$ over 2000 replications. Critical values are size-adjusted: each test is compared with the empirical 95th percentile of its own $|t|$ under $b = 0$. Size adjustment is what makes the comparison meaningful, since an oversized test mechanically has more raw power and a raw power curve would merely restate the size distortion.

6 Results

6.1 Conventional inference

Figure 1 shows conventional rejection frequencies. Equal positive persistence generates plateaus above 5%, and the plateau rises with $\phi_x\phi_y$. The unequal positive case $(0.9, 0.5)$ also over-rejects, but less severely than $(0.9, 0.9)$. The opposite-sign case $(0.8, -0.8)$ under-rejects sharply and approaches its theoretical rejection probability near zero. The random-walk rejection frequency continues toward one rather than settling at a stationary plateau, reaching 97.0% at $T = 5000$.

Most instructive is the AR/MA design. Both of its series are serially correlated — the AR component has $\rho_x(1) = 0.50$ — yet its rejection frequency sits on the nominal line, at 6.0% at $T = 5000$. Serial correlation alone is therefore not a diagnosis; it is the sign pattern of the two autocorrelation functions, through λ, that matters.

Table 1: Conventional rejection frequencies and stationary asymptotic benchmarks. Entries are proportions; the maximum Monte Carlo standard error is 1.12 percentage points. The I(1) row has no stationary benchmark.

DGP	λ	Asymptotic	$T = 50$	$T = 500$	$T = 5000$
iid	1.00	0.050	0.054	0.050	0.049
(0.5,0.5)	1.67	0.129	0.112	0.138	0.126
(0.8,0.8)	4.56	0.358	0.298	0.380	0.348
(0.9,0.9)	9.53	0.525	0.463	0.505	0.514
(0.95,0.95)	19.51	0.657	0.564	0.650	0.659
(0.9,0.5)	2.64	0.227	0.206	0.220	0.216
(0.8,-0.8)	0.22	2.9e-05	0.001	0.000	0.000
AR/MA lambda=1	1.00	0.050	0.041	0.043	0.060
I(1)	—	—	0.683	0.893	0.970

Table 1 confirms the theoretical classification. At $T = 5000$, the conventional rejection rates are 12.6% for $(0.5, 0.5)$, 34.8% for $(0.8, 0.8)$, 51.4% for $(0.9, 0.9)$, and 66.0% for $(0.95, 0.95)$, against asymptotic benchmarks of 12.9%, 35.8%, 52.5% and 65.7% respectively. The unequal positive case $(0.9, 0.5)$ rejects 21.6% of the time. For $(0.8, -0.8)$ the theoretical rejection probability is 0.0029%, so

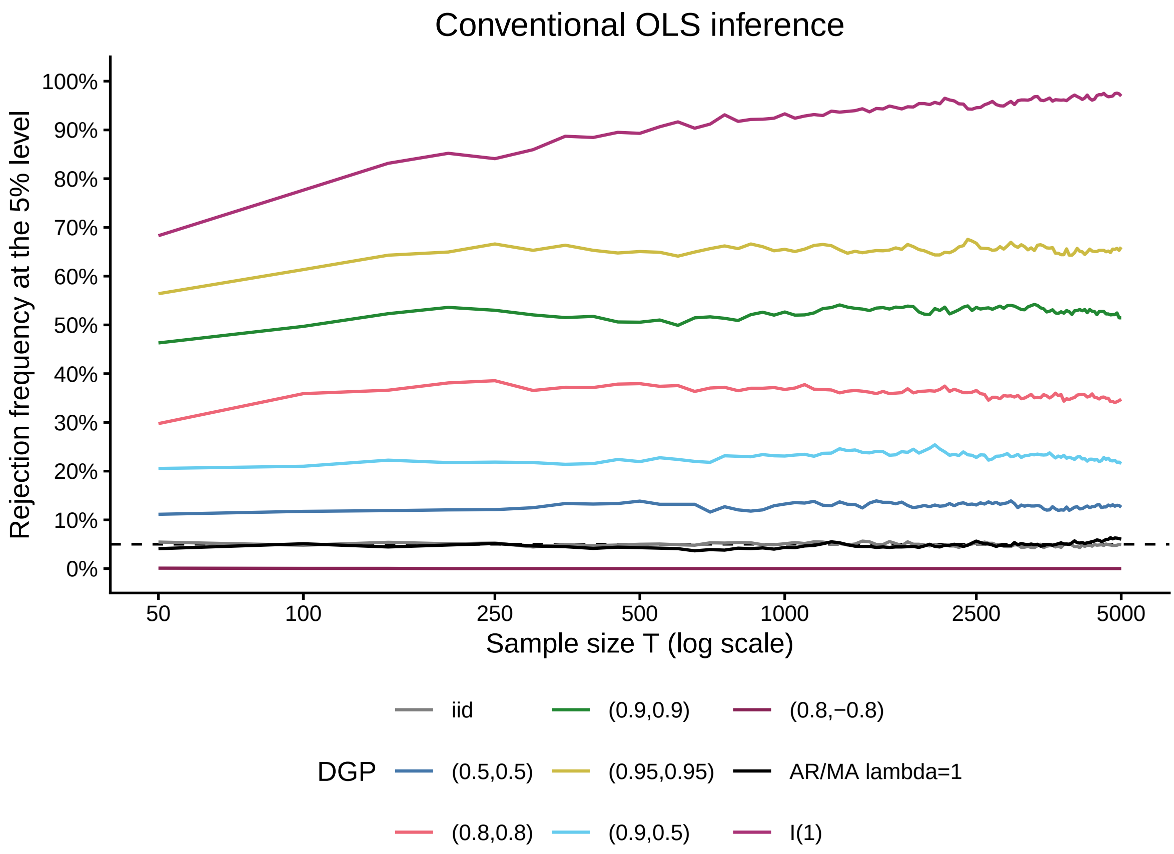

Figure 1: Conventional OLS rejection frequencies. Labels report the two AR coefficients where applicable; 'AR/MA lambda=1' is the AR(1)/MA(2) design for which lambda equals one exactly. The dashed line is the nominal 5% level. Note the logarithmic sample-size axis.

among 2000 replications one expects essentially none; the simulated frequency at $T = 5000$ is 0.0%. The i.i.d. and AR/MA frequencies fluctuate around the exact 5% benchmark rather than equalling 5% mechanically at every sample size.

6.2 HAC inference

Figure 2 compares conventional and HAC rejection frequencies for the stationary DGPs. At moderate and large T, HAC inference moves rejection rates toward 5% in every serially correlated case, including both the over-rejecting and the under-rejecting ones. At $T = 50$, however, the correction is dominated by long-run variance estimation noise and can be worse than no correction at all: for $(0.5, 0.5)$ the conventional test rejects 11.2% while HAC rejects 12.7%, and even the i.i.d. benchmark, where there is nothing to correct, rejects 11.5% under HAC. Convergence is slowest for $(0.95, 0.95)$, whose HAC rejection rate is still 10.7% at $T = 5000$. This is a finite-sample bandwidth and long-run variance estimation problem, not evidence against Proposition 10.

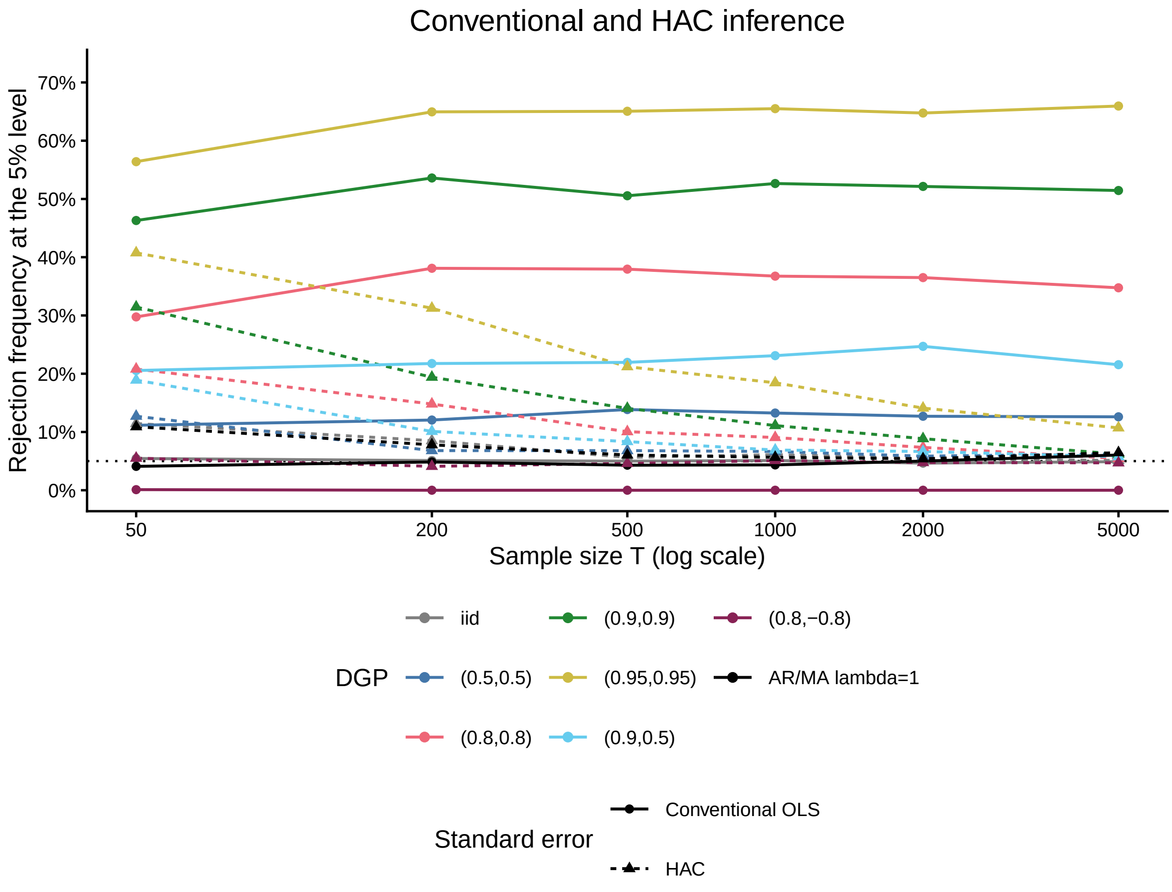

Figure 2: Conventional and Bartlett-HAC rejection frequencies for the stationary DGPs at selected sample sizes. The dotted line is the nominal 5% level. The bandwidth is floor(2 T^(1/3)).

Table 2: Conventional and HAC rejection frequencies at the endpoints. Entries are proportions; maximum Monte Carlo standard error 1.12 percentage points.

DGP	OLS $T = 50$	HAC $T = 50$	OLS $T = 5000$	HAC $T = 5000$
iid	0.054	0.114	0.049	0.051
(0.5,0.5)	0.112	0.127	0.126	0.062
(0.8,0.8)	0.298	0.208	0.348	0.054

Table 2: Conventional and HAC rejection frequencies at the endpoints. Entries are proportions; maximum Monte Carlo standard error 1.12 percentage points.

DGP	OLS $T = 50$	HAC $T = 50$	OLS $T = 5000$	HAC $T = 5000$
(0.9,0.9)	0.463	0.314	0.514	0.062
(0.95,0.95)	0.564	0.408	0.659	0.107
(0.9,0.5)	0.206	0.189	0.216	0.058
(0.8,-0.8)	0.001	0.055	0.000	0.047
AR/MA lambda=1	0.041	0.109	0.060	0.064

The HAC results caution against describing robust inference as an automatic finite-sample cure. At small T, estimating a long-run variance is itself noisy. By $T = 5000$ the HAC frequencies are close to 5% for every stationary DGP except the most persistent equal-coefficient case. Prewhitening or fixed-b critical values would be the natural next step for that case.

As in the local-power design below (Section 6.3), the rejection frequencies above characterise only the tail mass of the sampling distribution; a Jarque–Bera test on the same t-statistics, pooled over the 2000 replications behind Figure 2 at each DGP and sample size, checks its shape. At $T = 50$ normality is decisively rejected for the HAC statistic in every DGP examined, including the i.i.d. benchmark (statistic 89.72, $p = 0.0000$) and $(0.95, 0.95)$ (statistic 263.50, $p = 0.0000$); in the most persistent case even the conventional statistic is not immune (statistic 37.58, $p = 0.0000$). By $T = 5000$ the conventional statistic is normal in shape for every stationary DGP, and the only case in which HAC still shows a detectable departure is $(0.95, 0.95)$ (statistic 9.31, $p = 0.01$) – the same DGP already flagged above as the slowest to converge in rejection frequency. The two diagnostics therefore agree: what looks like a scale problem in Figure 2 is, at small T, also a shape problem, and both resolve together as T grows, except in the one design where persistence is highest.

6.3 Size-adjusted local power

Robust inference is often described as trading size for power. Figure 3 quantifies the trade for the $(0.8, 0.8)$ design, and the answer is milder than the folklore suggests. Once each test is compared with its own size-adjusted critical value, the two power curves are close: at $T = 250$ and $b = 4$ the conventional test has size-adjusted power 51.4% against 45.6% for HAC, and at $T = 1000$ the figures are 46.8% and 44.3%. The cost of HAC, properly measured, is the modest loss from estimating the long-run variance rather than knowing it — not the dramatic sacrifice that raw power comparisons suggest.

The size-adjusted critical values are themselves worth reporting, because they test Proposition 5 directly. For the $(0.8, 0.8)$ design, $\lambda = 4.556$, so the conventional two-sided 5% critical value should converge to $1.96\sqrt{\lambda} = 4.18$ rather than to 1.96. The simulated values are 4.03 at $T = 250$ and 4.16 at $T = 1000$, and they do not shrink with T — the distortion is a feature of the limit distribution, not a small-sample artefact. The corresponding HAC critical values, 2.50 and 2.21, do decline toward 1.96, as Proposition 10 requires.

The critical-value comparison above checks only the scale of the limit distribution; a Jarque–Bera test on the same null t-statistics checks its shape. At $T = 250$ the test does not reject normality for either method (statistic 0.55, $p = 0.76$ for the conventional t-ratio; 3.62, $p = 0.16$ for HAC), and the same is true at $T = 1000$ ($p = 0.29$ and $p = 0.16$ respectively). The limit is therefore normal in shape, not merely correctly scaled once λ is accounted for.

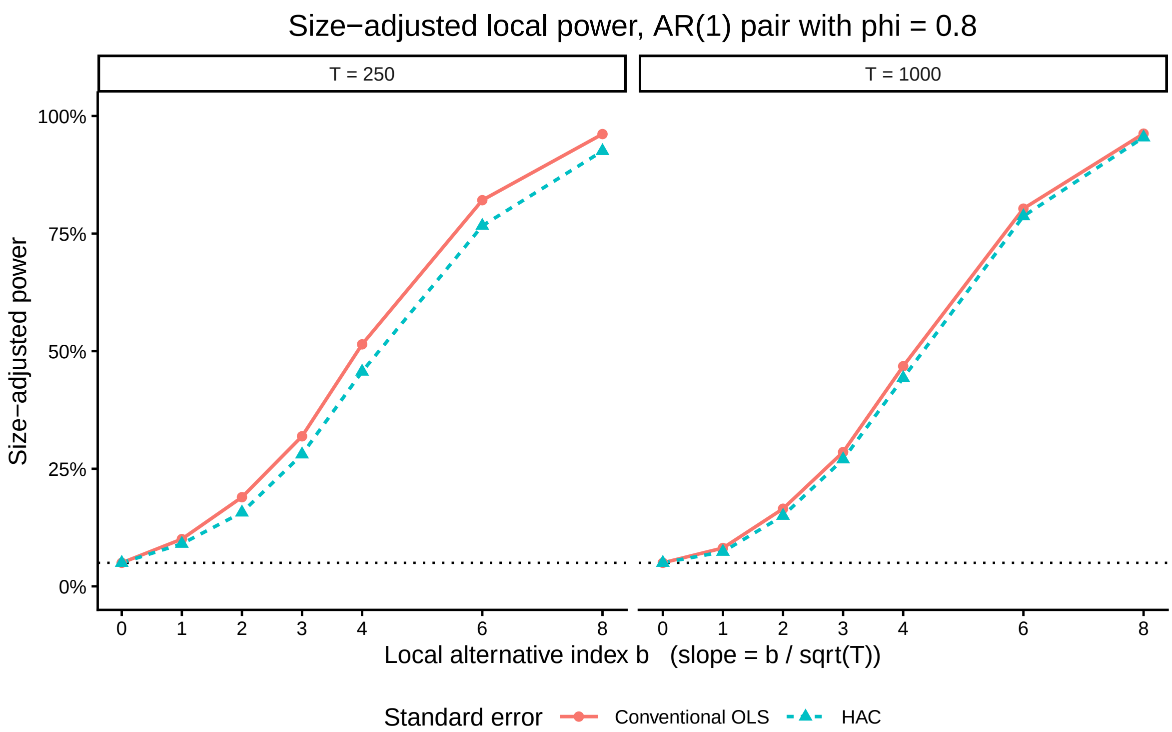

Figure 3: Size-adjusted local power against the alternative slope b / sqrt(T), for independent AR(1) series with phi = 0.8. Each test uses its own empirical 95th percentile under b = 0 as critical value, so all curves pass through 5% at b = 0 by construction.

6.4 Dependent zero-projection series and common volatility

Figure 4 relaxes full independence. Although x_t and y_t share the same stochastic scale, their population projection slope is zero because their innovations are independent, and their score is exactly serially uncorrelated. The conventional rejection frequency reaches 13.9% at $T = 5000$, against the 12.7% asymptotic value implied by $\lambda = \kappa = e^{0.5}$. HC0 rejects 5.7% and HAC 6.0% at the same sample size. This is the counterexample of Remark 7 in action: a size distortion generated entirely by contemporaneous second-moment dependence, with no serial autocovariance in the score whatsoever. HC suffices; HAC also works but pays for estimating autocovariances that are zero.

6.5 Local-to-unity sequences

Figure 5 shows that fixed-parameter stationary conclusions are not uniform near the unit-root boundary. As T rises, $\phi_T = 1 - 5/T$ moves closer to one. Conventional rejection rises from 47.9% at $T = 50$ to 94.3% at $T = 5000$ rather than approaching a fixed stationary plateau, and HAC deteriorates along the same sequence, from 33.7% to 71.4%. Each finite-T DGP is stationary, yet the sequence behaves increasingly like the unit-root benchmark: at $T = 5000$ the local-to-unity conventional rejection frequency of 94.3% is close to the I(1) benchmark's 97.0%. As noted in Section 4.8, the theoretical rejection probability tends to one along this sequence, exactly as under a unit root. A unit-root or near-unit-root diagnostic is therefore complementary to robust covariance estimation, not substitutable for it.

6.6 Distributed-lag and Granger-causality tests

The null in Figure 6 is that two lags of x jointly have no predictive content for y. Because the two AR processes are generated independently, this null is true — and, as explained in Section 4.10, it

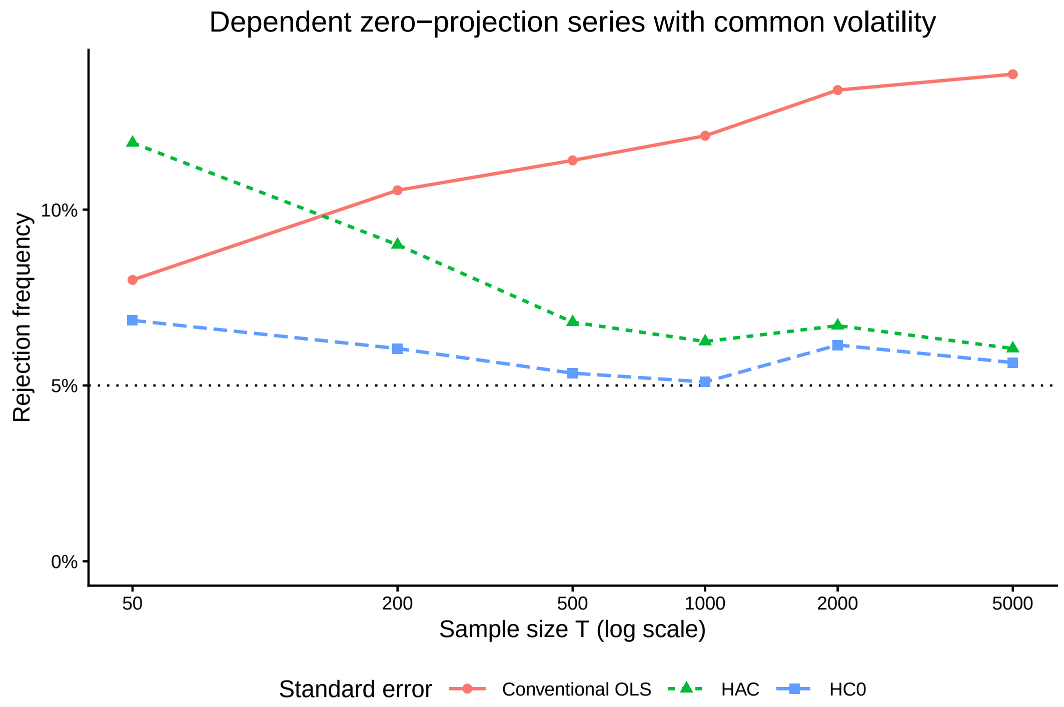

Figure 4: Rejection frequencies for dependent zero-projection series sharing a persistent stochastic volatility. The dotted line is the nominal 5% level.

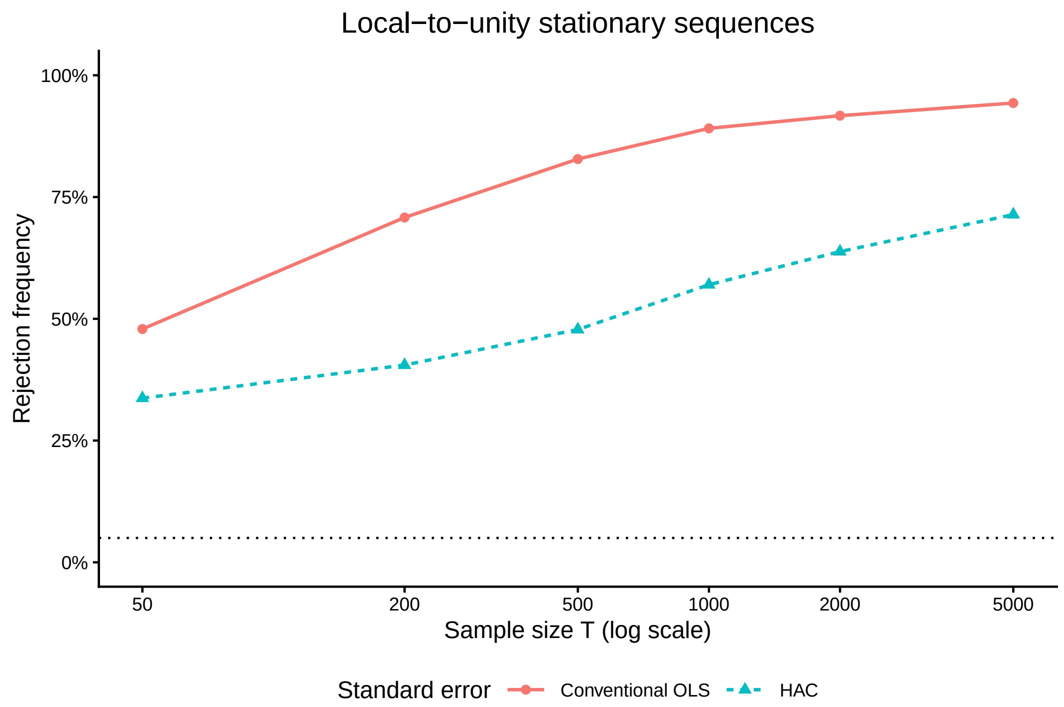

Figure 5: Rejection frequencies for local-to-unity AR(1) sequences with phi_T = 1 - c/T. Each finite-sample DGP is stationary, but the parameter approaches unity with T.

is true in *both* specifications, since the omitted y_{t-1} is orthogonal to the included x lags. There is no coefficient bias anywhere in this experiment.

When the model includes y_{t-1}, the conventional joint Wald rejection frequency is 5.6% at $T = 100$ and 6.5% at $T = 2000$. HAC is more distorted in the smallest samples, at 12.6% at $T = 100$, but declines toward nominal size.

When y_{t-1} is omitted, the conventional test rejects 29.8% at $T = 100$ and 31.6% at $T = 2000$, creating persistent false evidence of Granger causality that does not diminish with sample size. HAC reduces this to 6.3% at $T = 2000$. Readers should check whether that figure has settled or is still declining across the last two sample sizes in the plot: convergence to nominal size is guaranteed asymptotically but can be slow when the induced error is as persistent as it is here.

The comparison separates two remedies. Including y's dynamics correctly specifies the conditional mean; HAC addresses the covariance of the remaining score. Here only the second problem is present, and HAC accordingly does most of the work. In applications where the omitted dynamics are correlated with the regressors, the first problem is present too and HAC cannot help. Applied work generally needs both.

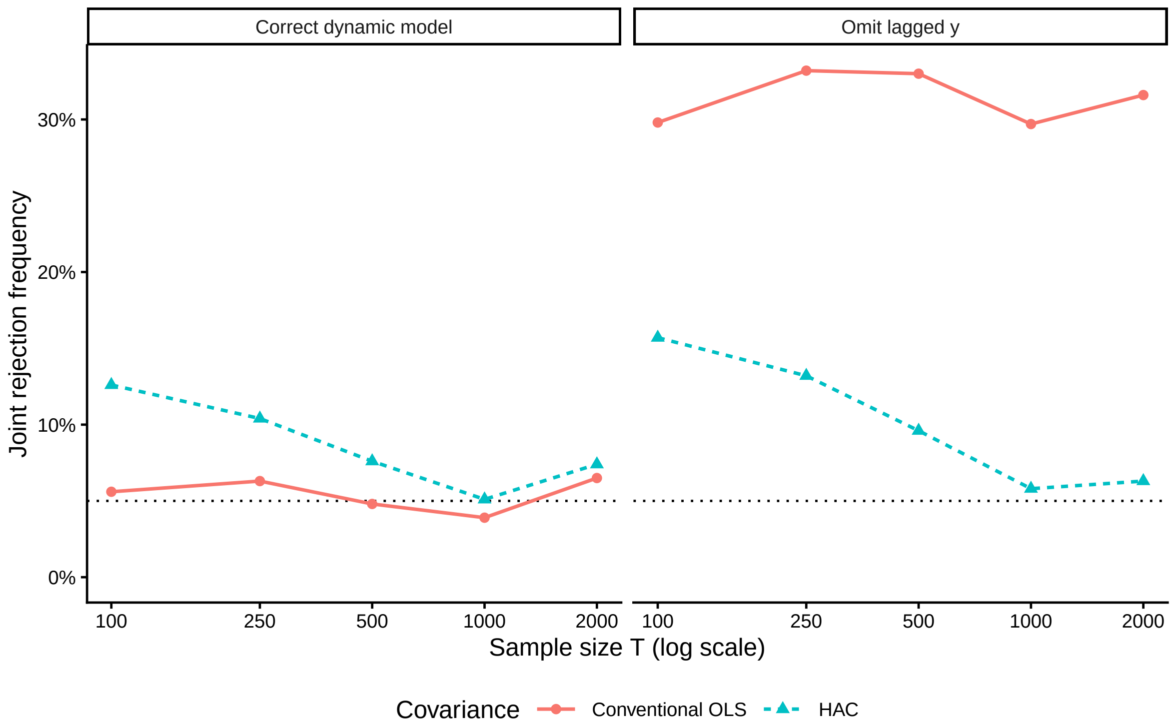

Figure 6: Joint Wald tests of no Granger causality from x to y. The correct dynamic model includes lagged y; the misspecified model omits it. The dotted line is the nominal 5% level.

Table 3: Selected rejection frequencies from the scope-extension experiments. HC0 is reported only where contemporaneous heteroskedasticity is the operative distortion.

Experiment	Conventional	HC0	HAC
Common volatility, $T = 5000$	0.138	0.056	0.060
Local-to-unity, $T = 5000$	0.943	—	0.714
Granger, correct dynamics, $T = 2000$	0.065	—	0.074
Granger, omitted dynamics, $T = 2000$	0.316	—	0.063

7 Discussion and Conclusion

The theory and simulation together support a more precise answer than the assertion that spurious regression either disappears or necessarily persists under stationarity.

First, joint short-memory stationarity and projection orthogonality make OLS consistent for the population projection slope under the stated assumptions. Full independence is unnecessary. The common-volatility experiment shows that dependent series can have a zero slope while still invalidating the conventional homoskedastic standard error. This sharply distinguishes the stationary setup from the independent random-walk regression, in which the slope has a nondegenerate limit and the conventional t-ratio diverges.

Second, consistency of the coefficient does not validate its conventional standard error. The limiting conventional test depends on λ, which by Lemma 6 has two independent sources: a contemporaneous component κ and a serial component. Correct asymptotic size requires the two together to sum to one. Neither the presence nor the absence of serial correlation settles the matter, as the AR/MA design and the common-volatility design demonstrate from opposite directions. The equal-positive AR(1) case is economically important, but it is an example rather than a universal law.

Third, increasing the sample size removes transient finite-sample effects but does not eliminate a non-nominal conventional plateau. The size-adjusted critical values in Section 6.3 make this concrete: they converge to $1.96\sqrt{\lambda}$, not to 1.96, and they do not shrink as T grows.

Fourth, HAC inference is the appropriate asymptotic remedy when its additional consistency conditions hold. The simulation confirms movement toward nominal size and shows that the power cost, correctly measured against size-adjusted critical values, is modest. It also shows that high persistence makes convergence slow and bandwidth choice consequential, and that at very small T the correction can be worse than no correction. Prewhitening and fixed-b critical values are the standard responses.

Fifth, fixed-parameter stationary asymptotics can be misleading near the unit-root boundary. The local-to-unity experiment produces rejection probabilities tending toward one even though every finite-sample autoregressive coefficient is below one. Stationarity is therefore not a uniform guarantee over empirically plausible parameter sequences.

Finally, robust inference cannot replace dynamic specification. Distributed-lag and Granger-causality tests require an information set rich enough to capture the dependent variable's own dynamics. Omitting those dynamics leaves serial dependence in the error and can create persistent false predictive relationships. In the design used here that failure is purely one of covariance estimation and HAC largely repairs it; when the omitted dynamics are correlated with the regressors, the coefficients are biased too and no covariance estimator can help.

The practical conclusion is conditional but clear: short-memory stationarity and projection orthogonality protect slope consistency, not conventional inference. Researchers should diagnose dependence and heteroskedasticity in the regression score, assess proximity to the unit-root boundary, specify relevant dynamics, and use a defensible covariance estimator rather than assume that either stationarity or a large sample makes the usual OLS t-ratio or Granger-causality Wald test reliable.

Exercises

Exercise 1 (The product, not the persistence)**.** Two independent AR(1) series have $\phi_x = 0.9$ and $\phi_y = -0.9$. Compute λ and the asymptotic rejection probability of a nominal 5% two-sided t-test. Compare with $\phi_y = +0.9$. Explain in one sentence why persistence in each series separately is not the relevant statistic.

Exercise 2 (Constructing correct size)**.** Using $\lambda = \sum_k \rho_x(k)\rho_y(k)$, find a value of θ_2 such that an AR(1) regressor with $\phi = 0.7$ paired with an invertible MA(2) yields $\lambda = 1$. Verify invertibility. Then modify `lambda1_phi` and `lambda1_theta2` in the simulation script and confirm the rejection frequency numerically.

Exercise 3 (Where does the distortion come from?)**.** For the common-volatility design of Section 4.7, verify that $\text{Cov}(w_t, w_{t+k}) = 0$ for all $k \neq 0$, and that $\kappa = \mathbb{E}(s_t^4)/\{\mathbb{E}(s_t^2)\}^2$. Explain why HC0 is sufficient here and what would change if the innovations ε_t were themselves serially correlated.

Exercise 4 (Reading a size-adjusted critical value)**.** Section 6.3 reports a size-adjusted conventional critical value near $1.96\sqrt{\lambda}$ that does not shrink with T. Explain why this is a sharper check on Proposition 5 than the rejection frequency, and what you would expect to see if Proposition 5 were false.

Exercise 5 (Bandwidth sensitivity)**.** Replace `hac_bandwidth` in the simulation script with the Newey–West rule of thumb $\lfloor 4(T/100)^{2/9} \rfloor$ and re-run the HAC comparison. At $T = 5000$ this changes the bandwidth from 34 to 9. Predict, before running, whether the $(0.95, 0.95)$ distortion will get better or worse, and explain the result you obtain.

Exercise 6 (Diagnosis in practice)**.** You estimate a regression on quarterly data with $T = 120$ and obtain a t-statistic of 2.6. List the diagnostics you would run, in order, before reporting this as evidence of a relationship. For each, state what result would change your conclusion.

Bibliography

The list below includes both the works cited in the text and general background reading. Students meeting this material for the first time will find Hamilton (1994) the most useful single reference for the underlying time-series asymptotics, and Engle (1982) the origin of the conditional-heteroskedasticity models that motivate the common-volatility design of Section 4.7.

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Computational Appendix

The complete, executable simulation is maintained in `scripts/spurious_stationary_sim.R`. It depends only on `ggplot2` and `scales`; the Bartlett HAC estimator used throughout is implemented in the script itself so that a single estimator produces every robust number reported above. Running

```
Rscript scripts/spurious_stationary_sim.R
```

recreates all figures, the CSV result files, and a `sessionInfo.txt` recording the package versions used. Setting `SPURIOUS_QUICK=1` in the environment runs a fast, noisier pass suitable for drafting; `SPURIOUS_OUTPUT_DIR` redirects the output location. Each experiment is seeded independently, so changing the replication count of one does not silently alter the results of another. The Quarto document sources that same script, so the handout and the standalone computation share one implementation, and every number in the text is read from the simulation output at render time.